puting are a far cry from the kind that would be needed in order to tackle the awesome power bursting from the proper concept that is Quantum Computing. But because no concept would ever be allowed to run loose in today's world, counter-ideas for the force of Quantum Computing are topics of hot debate, and of course, research.

It is well known that unless we stumble upon another hitherto unforeseen blockage in nature (we're looking at you Heisenberg Principle), Quantum Computers would make it through. That day might not be today, and it might not be tomorrow, but it will be here someday. And if a power exists, we need to consider that it will be used for at least some sort of evil, especially when it holds such never-seen-before capabilities. Consider this: the moment the day comes that a Quantum Computer is announced to the world, RSA is dead, DSA is dead; hyperbolic, elliptic, hyperelliptic, and every other curve based encryption technique is dead. And if we really want to make sure that every form of data protection technique known to man is not dead with all of that, we need to find remedies. And we need to find them now.

There are some ideas in that space in existence, though. These include:

1. Hash based cryptograpy

If you have any background in computer science, you might have heard of hash tables. And if you have some serious background, you probably love them, like some of us. To hear that these little things are quantum proof in the broad sense is a huge relief. Yes, folks, hashing would survive even in the era of quantum computers, because the idea here is that we shouldn't be solely reliant on a secret trapdoor functions to produce our keys for us. Then again, if random number based keys are the only hope for the future, then it might not be that great after all.